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THE HONG KONG POLYTECHNIC UNIVERSITY

DEPARTMENT OF LAND SURVEYING AND GEO-INFORMATICS

Programme	**:**	34414 BSc (Hons) in Geomatics 34478 BSc (Hons) in Land Surveying and Geo-Informatics 34379 HD in Land Surveying and Geo-Informatics			
Subject Title	**:**	Geospatial Database Management and Design			
Subject Code	**:**	LSGI3245	**Session**	**:**	Semester 1, 2021/22
Date	**:**	13 December 2021	**Time**	**:**	12:30 - 14:30
Time Allowed	**:**	2 hours	**Subject Examiner(s)**	**:**	Dr. Yang Xu (LSGI)

This question paper has a total of 5 pages.
(Some pages may be intentionally omitted.)

Instructions to Candidates : This paper has **TWO** sections, Section A and B.

Section A: **COMPULSORY**.
Section B: Answer any **ONE** question.

Questions carry marks as indicated.

Total marks = 100.

Available from Invigilator : Nil

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Section A COMPULSORY

Question 1 **(True/False)**

Question 2 **(Multiple Choice question)**

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Question 3

(a) Name at least FOUR drawbacks of using traditional file-based systems over database management systems. Use 1-2 sentences to explain each advantage that you list. (10 marks)

(b) What is the definition of a database transaction? Use your own words to explain it and also provide a real-world example. After that, explain the difference between the following two concepts:

- A database transaction needs to be atomic.
- An attribute field in a relation is usually atomic.

(10 marks)

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Question 4

Explain the concept of Z-Curve as a Geo-Hashing technique for spatial database. Use an example to explain how Z-order is calculated.

(10 marks)

Question 5

Given the following two tables:

The **staff** table.

	name [PK] character varying (10)	dept character varying (10)	role character varying (10)
1	James	Physics	Professor
2	Clark	Music	Lecturer

The **salary** table.

	dept character varying	role character varying	salary numeric
1	Music	Professor	20000
2	Music	Lecturer	15000
3	Music	PhD Student	10000
4	Physics	Professor	18000
5	Physics	Lecturer	16000
6	Physics	PhD Student	12000

Please write down your answer, in the form of a relation, for the following SQL expressions:

from staff natural join salary; (5 marks)

(b) select *
from staff join salary on people.dept = salary.dept; (5 marks)

(c) select *
from staff join salary using (role); (5 marks)

(d) select *
from staff natural right outer join salary; (5 marks)

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Section B: Answer any ONE question.

Question 6

Use 1-2 paragraphs to explain your understanding of the following concepts:

- Logical **schema** and **instance** of a database
- Relation **schema** and **instance**

(10 marks)

Question 7

Please create an example (e.g. a table with columns/rows) to explain the concepts of **primary key**, **candidate key(s)**, and **superkeys**, and their relationships.

(10 marks)

- END OF PAPER -